

Matrix

• Matrix

A matrix is a rectangular array of numbers, symbols or expressions arranged in rows and columns within a pair of brackets.

• Determinant

A determinant is a scalar that is computed the elements of a square matrix.

• Row matrix

$$[1 \quad 2 \quad 3]$$

• Main diagonal

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \quad n \times n$$

• Column

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$a_{11}, a_{22}, \dots, a_{nn}$$

$$\text{Trace} : a_{11} + a_{22} + \dots + a_{nn}$$

• Diagonal matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

• Scalar Matrix

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Order

row $\times$ column.

by

400 minutes (154)

If $|A| = 0$ then A is called a singular Matrix

If $|A| \neq 0$

[153]

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$\therefore A$ is non-singular matrix.

* Show that

is singular.

$$|A| = \begin{vmatrix} +1 & -2 & +3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 0$$

* Prove that,

$$A = \begin{vmatrix} 13 & 16 & 19 \\ 14 & 17 & 20 \\ 15 & 18 & 21 \end{vmatrix}; \text{ is singular matrix.}$$

$$\begin{aligned} |A| &= 13(357 - 360) - 16(294 - 300) + 19(252 - 255) \\ &= 13 \times (-3) + 96 + -57 \\ &= 0. \end{aligned}$$

• Transpose of a matrix

$$A = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$$

$$A' / A^t / A^T = \begin{vmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{vmatrix}$$

* Imaginary number : $\sqrt{-1} = \sqrt{i^2} = i$

* complex number : A number of the form $x+iy$ is called a complex number, where x, y are real number and $i^2 = -1$.
 $i, -i, 1+2i, 3-4i$

* Conjugate of complex number

$$z = x + iy$$

$$\bar{z} = x - iy$$

$$z = -x - iy$$

$$\bar{z} = -x + iy$$

• Hermitian and skew-Hermitian matrix:

Let A be a complex number

If $(A)^T = A$ then A is called a Hermitian matrix

If $(A)^T = -A$ then A is called a skew-Hermitian matrix

* Prove that,

$$A = \begin{bmatrix} 2 & 2-3i & 3+5i \\ 2+3i & 3 & i \\ 3-5i & -i & 5 \end{bmatrix}$$

is Hermitian matrix.

$$\bar{A} = \begin{bmatrix} 2 & 2+3i & 3-5i \\ 2-3i & 3 & -i \\ 3+5i & i & 5 \end{bmatrix}$$

$$(\bar{A})^T = \begin{bmatrix} 2 & 2+3i & 3-5i \\ 2-3i & 3 & -i \\ 3+5i & i & 5 \end{bmatrix} = \begin{bmatrix} 2 & 2-3i & 3+5i \\ 2+3i & 3 & i \\ 3-5i & -i & 5 \end{bmatrix} = A$$

Since $(\bar{A})^T = A$, A is Hermitian.

* Prove that,

$$A = \begin{vmatrix} 2 & 1+2i & 2+3i \\ 1-2i & 5 & 3+5i \\ 2-3i & 3-5i & 6 \end{vmatrix} \text{ is Hermitian matrix.}$$

$$\bar{A} = \begin{vmatrix} -2 & 1-2i & 2-3i \\ 1+2i & -5 & 3-5i \\ 2+3i & 3+5i & -6 \end{vmatrix}$$

$$(\bar{A})^T = \begin{vmatrix} -2 & 1-2i & 2-3i \\ 1+2i & -5 & 3-5i \\ 2+3i & 3+5i & -6 \end{vmatrix}$$

$$\textcircled{11} = \begin{vmatrix} 2 & 1+2i & 2+3i \\ 1-2i & 5 & 3+5i \\ 2-3i & 3-5i & 6 \end{vmatrix}$$

$$= A.$$

$\therefore$ Since $(\bar{A})^T = A$, A is Hermitian.

* Show that,

$$A = \begin{bmatrix} i & 2-3i & 4+5i \\ -2-3i & 0 & 2i \\ -4+5i & 2i & -3i \end{bmatrix} \text{ is skew Hermitian.}$$

$$A = \begin{bmatrix} i & 2-3i & 4+5i \\ -2-3i & 0 & 2i \\ -4+5i & 2i & -3i \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} -i & 2+3i & 4-5i \\ -2+3i & 0 & -2i \\ -4-5i & -2i & -3i \end{bmatrix}$$

$$(\bar{A})^T = \begin{bmatrix} -i & 2+3i & -4-5i \\ 2+3i & 0 & -2i \\ 4-5i & -2i & -3i \end{bmatrix}$$

Multiplying ① by -1 , we get

$$-A = \begin{vmatrix} -i & -2+3i & -4-5i \\ 2+3i & 0 & -2i \\ 4-5i & -2i & 3i \end{vmatrix} \quad \text{--- (ii)}$$

from ① & ② we get,

$$(\bar{A})^T = -A$$

$\therefore A$ is skew-Hermitian.

* If $H = \begin{vmatrix} 3 & 5+2i & -3 \\ 5-2i & 7 & 4i \\ -3 & -4i & 5 \end{vmatrix}$

show that, H is a Hermitian matrix

$$\Rightarrow \bar{A} = \begin{vmatrix} -3 & 5-2i & 3 \\ 5+2i & 7 & -4i \\ 3 & 4i & -5 \end{vmatrix}$$

$$A = (A)^T = \begin{vmatrix} -3 & 5-2i & 3 \\ 5+2i & -7 & -4i \\ 3 & 4i & -5 \end{vmatrix} = \begin{vmatrix} 3 & 5+2i & -3 \\ 5-2i & 7 & 4i \\ -3 & -4i & 5 \end{vmatrix} = A.$$

Since, $(A)^T = A$, A is Hermitian.

Again,

$$A = \begin{vmatrix} 3 & 5+2i & -3 \\ 5-2i & 7 & 4i \\ -3 & -4i & 5 \end{vmatrix}$$

$$\bar{A} = \begin{vmatrix} -3 & 5-2i & 3 \\ 5+2i & -7 & -4i \\ 3 & 4i & -5 \end{vmatrix}$$

$$(\bar{A})^T = \begin{vmatrix} -3 & 5-2i & 3 \\ 5+2i & -7 & -4i \\ 3 & 4i & -5 \end{vmatrix} \quad \text{--- (ii)}$$

Multiplying (i) by -1 , we get,

$$-A = \begin{vmatrix} -3 & 5-2i & 3 \\ 5+2i & -7 & -4i \\ 3 & 4i & -5 \end{vmatrix} \quad \text{--- (iii)}$$

from (ii) & (iii) we get,

$$(\bar{A})^T = -A$$

$\therefore A$ is a skew-Hermitian.

- Idempotent matrix

Let A be a square matrix

If $A^2 = A$ then A is called an idempotent matrix.

- Involutory matrix

Let A be a square matrix

If $A^2 = I$ then A is called an involutory matrix.

* Show that,

$$A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \text{ is an idempotent matrix.}$$

We have,

$$A^2 = A \cdot A$$

$$= \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 4+2-4 & -4-6+8 & -8-8+12 \\ -2-3+4 & 2+9-8 & 4+12-12 \\ 2+2-3 & -2-6+6 & -4-8+9 \end{bmatrix}$$

$$= \begin{vmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{vmatrix}$$

$$= A$$

Since, $A^2 = A$, A is an idempotent matrix.

* Show that,

$$A = \begin{vmatrix} -1 & 1 & -1 \\ 3 & -3 & 3 \\ 5 & -5 & 5 \end{vmatrix}$$

is an idempotent matrix.

We have,

$$A^2 = A \cdot A$$

$$= \begin{vmatrix} -1 & 1 & -1 \\ 3 & -3 & 3 \\ 5 & -5 & 5 \end{vmatrix} \begin{vmatrix} -1 & 1 & -1 \\ 3 & -3 & 3 \\ 5 & -5 & 5 \end{vmatrix}$$

$$= \begin{vmatrix} 1+3-5 & -1-3+5 & 1+3-5 \\ -3-9+15 & 3+9-15 & -3-9+15 \\ -5-15+25 & 5+15-25 & -5-15+25 \end{vmatrix}$$

$$= \begin{bmatrix} -1 & 1 & -1 \\ 3 & -3 & 3 \\ 5 & -5 & 5 \end{bmatrix}$$

$$= A.$$

Since,

$$A^2 = A,$$

A is an idempotent matrix.

* Show that,

$$A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$$

is an involutory matrix.

We have,

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

Now,

$$A^2 = A \cdot A$$

$$= \begin{vmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{vmatrix} \begin{vmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 25 - 24 + 0 & 40 - 40 + 0 & 0 + 0 + 0 \\ -15 + 15 + 0 & -24 + 25 + 0 & 0 + 0 + 0 \\ -5 + 6 - 1 & -8 + 10 - 2 & 0 + 0 + 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= I$$

$$\therefore A^2 = I$$

$\therefore A$ is an involutory matrix.

• Orthogonal matrix

A square matrix A is called orthogonal if

$$AA^T = A^T A = I,$$

where I is the unit matrix

* Show that,

$$A = \frac{1}{9} \begin{vmatrix} 1 & 8 & -4 \\ 4 & -4 & -7 \\ 8 & 1 & 4 \end{vmatrix} \text{ is orthogonal}$$

$$\Rightarrow A^T = \frac{1}{9} \begin{vmatrix} 1 & 4 & 8 \\ 8 & -4 & 1 \\ -4 & -7 & 4 \end{vmatrix}$$

$$\Rightarrow AA^T = \frac{1}{9} \begin{vmatrix} 1 & 8 & -4 \\ 4 & -4 & -7 \\ 8 & 1 & 4 \end{vmatrix} \cdot \frac{1}{9} \begin{vmatrix} 1 & 4 & 8 \\ 8 & -4 & 1 \\ -4 & -7 & 4 \end{vmatrix}$$

$$= \frac{1}{81} \begin{vmatrix} 1 & 8 & -4 \\ 4 & -4 & -7 \\ 8 & 1 & 4 \end{vmatrix} \begin{vmatrix} 1 & 4 & 8 \\ 8 & -4 & 1 \\ -4 & -7 & 4 \end{vmatrix}$$

$$= \frac{1}{81} \begin{vmatrix} 1+64+16 & 4-32+28 & 8+8-16 \\ 4-32+28 & 16+16+49 & 32-4-28 \\ 8+8-16 & 32-4-28 & 64+1+16 \end{vmatrix}$$

$$= \frac{1}{81} \begin{vmatrix} 81 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 81 \end{vmatrix}$$

$$= \frac{1}{81} \times 81 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = I$$

Since $AA^T = I$.

So A is orthogonal.

• Show that,

$$i) A = \frac{1}{3} \begin{vmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{vmatrix}$$

$$ii) B = \begin{vmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

are orthogonal matrices.

① Given,

$$A = \begin{vmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{vmatrix}$$

$$\Rightarrow A^T = \begin{vmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{vmatrix}$$

therefore,

$$A A^T = \begin{vmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{vmatrix} \begin{vmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 1+4+4 & 2+2-4 & -2+4-2 \\ 2+2-4 & 4+1+4 & -4+2+2 \\ -2+4-2 & -4+2+2 & 4+4+1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = I$$

Since,

$AA^T = I$, So A is Orthogonal

②

Given,

$$B = \begin{vmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$\Rightarrow B^T = \begin{vmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

therefore,

P.T.O

$$BB^T = \begin{vmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} \cos^2\alpha + \sin^2\alpha + 0 & \cos\alpha\sin\alpha - \sin\alpha\cos\alpha & 0+0+0 \\ \sin\alpha\cos\alpha - \sin\alpha\cos\alpha + 0 & \sin^2\alpha + \cos^2\alpha + 0 & 0+0+0 \\ 0+0+0 & 0+0+1 & 0+0+1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= I$$

Since,

$BB^T = I$, So B is Orthogonal.

* Check where it is singular

$$\begin{vmatrix} 7 & 2 & f \\ 2 & 4 & e \\ 3 & 5 & 1 \end{vmatrix},$$

where e and f are the digits of your own metric number.

$\Rightarrow$ for example,

$$e = 2$$

$$f = 3$$

We have,

$$\begin{vmatrix} 7 & 2 & 3 \\ 2 & 4 & 2 \\ 3 & 5 & 1 \end{vmatrix}$$

$$= 7(4 - 10) - 2(2 - 6) + 3(10 - 12)$$

$$= -42 + 8 - 6$$

$$= -40 \neq 0$$

So the given matrix is non-singular

* Construct 4×4 matrix having,

$$A = (a_{ij}) = 4i + j, \quad i > j$$

$$= 0, \quad i = j$$

$$= 3i - j, \quad i < j$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 4 & 0 & -1 \\ 9 & 0 & 3 & 2 \\ 13 & 14 & 0 & 5 \\ 17 & 18 & 19 & 0 \end{bmatrix}$$

• Unitary matrix

A square matrix A is called unitary if

$$AA^* = A^*A = I$$

where, $A^* = (\bar{A})^T$

* prove that, $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ is unitary.

$$\Rightarrow \bar{A} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1-i \\ 1+i & -1 \end{bmatrix}$$

$$\Rightarrow A^* = (\bar{A})^T = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

$$\Rightarrow AA^* = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 0 & 1+i \\ 1-i & 0 & -1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{vmatrix} 3 & 0 \\ 0 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = I.$$

$\therefore A$ is unitary

Q * If $A = [a_{ij}]$, where $a_{ij} = \begin{cases} 0, & i \neq j \\ c, & i = j \end{cases}$ construct a 3×3 order matrix and identify the type of the matrix. Also test the matrix A is orthogonal or not.

⇒ for example,

Here $c = 2 + 1 = 3$

$$a_{ij} = \begin{cases} 0, & i \neq j \\ 3, & i = j \end{cases}$$

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

∴ A is a scalar matrix

Again,

$$AA^T = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 9+0+0 & 0+0+0 & 0+0+0 \\ 0+0+0 & 0+9+0 & 0+0+0 \\ 0+0+0 & 0+0+0 & 0+0+9 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

• Adjoint and Inverse

→ Inverse of a matrix : A square matrix A is said to be invertible if there exists a matrix B such that $AB = BA$.

* formula : $A^{-1} = \frac{1}{|A|} \text{adj. } A$

$\text{adj. } A = \text{adjoint matrix}$

* Methods for finding inverse matrix :

- i) Elementary row transformation.
- ii) Adjoint matrix
- iii) Linear system of equations
- iv) By Cayley Hamilton theorem
- v) Gauss Elimination method
- vi) " Jordan "
- vii) Cramer's method
- viii) Doolittle "
- ix) Choleski's "
- x) Escalator "
- xi) Iterative method.

Q * find A^{-1} for $A = \begin{vmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{vmatrix}$ and also show that $A^3 = A^{-1}$.

$$\rightarrow |A| = \begin{vmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{vmatrix} = 3(1 - 4) - 3(4 - 0) + 4(-2 - 0) = 3(-3) - 3(4) + 4(-2) = -9 - 12 - 8 = -29 \neq 0$$

Since $|A| \neq 0$, so A^{-1} exists.

The co-factors of each element of $|A|$ are

$$A_{11} = (-1)^{1+1} \begin{vmatrix} -3 & 4 \\ -1 & 1 \end{vmatrix} = -3 + 4 = 1$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 2 & 4 \\ 0 & 1 \end{vmatrix} = -2$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} 2 & -3 \\ 0 & -1 \end{vmatrix} = -2$$

$$A_{21} = -1$$

$$A_{22} = 3$$

$$A_{23} = 3$$

$$A_{31} = 0$$

$$A_{32} = -4$$

$$A_{33} = -3$$

$$\text{adj. } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}^T$$

$$= \begin{bmatrix} 1 & -2 & -2 \\ -1 & 3 & 3 \\ 0 & -4 & -3 \end{bmatrix}^T$$

$$= \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{|A|} \text{adj. } A = \frac{1}{1} \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

Again, Show that, $AB = A^{-1}$.

Given,

$$A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

Now,

$$A^2 = A \cdot A$$

$$= \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 9-6+0 & -9+9-4 & 12-12+4 \\ 6-6+0 & -6+9-4 & 8-12+4 \\ 0-2+0 & 0+3-1 & 0-4+1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & 2 & -3 \end{bmatrix}$$

$$A \cdot A^{-1} = I$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$0+8-0$$

$$\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

$$12-12+4$$

$$8-12+4$$

$$0-4+1$$

$$A \cdot A^{-1} = I$$

Now,

$$A^3 = A^2 \cdot A$$

$$= \begin{bmatrix} 3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & 2 & -3 \end{bmatrix} \begin{bmatrix} 4 & 3 & -3 \\ 0 & 2 & -3 \\ -3 & 0 & 1 \end{bmatrix} = A$$

$$= \begin{bmatrix} 9-8+0 & -9+12-4 & 12-16+4 \\ 0-2+0 & 0+3+0 & 0-4+0 \\ -6+4+0 & +6-6+3 & -8+8-3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & -1 \\ -2 & 3 & -3 \\ -2 & 3 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & -1 \\ -2 & 3 & -3 \\ -2 & 3 & -3 \end{bmatrix} \xrightarrow{R_2 + 2R_1, R_3 + 2R_1} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_1 + R_2} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore A^3 = A^{-1}$$

showed :-

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Q * If $A = \begin{vmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{vmatrix}$ then find A^{-1} .

If $A = \begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{vmatrix}$ & $B = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{vmatrix}$

then find $A^{-1}B$ & AB^{-1} .

* Solution of linear equation : Matrix method :

g Solve with the help of matrix $\rightarrow$

$$x + 2y - z = 0$$

$$3x + 8y + 2z = 7$$

$$4x + 9y - z = 4$$

$\Rightarrow$ The given system can be written in matrix

form :

$$AX = B$$

where,

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 8 & 2 \\ 4 & 9 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 7 \\ 4 \end{bmatrix}$$

A^{-1}

$$|A| = 1 \neq 0$$

so A^{-1} exists.

Now,

$$A_{11} = -26$$

$$A_{21} = -7$$

$$A_{31} = 12$$

$$A_{12} = 11$$

$$A_{22} = 3$$

$$A_{32} = -5$$

$$A_{13} = -5$$

$$A_{23} = -1$$

$$A_{33} = 2$$

$$\text{adj} = \begin{vmatrix} -26 & 11 & -5 \\ -7 & 3 & -12 \\ 12 & -5 & 2 \end{vmatrix}^T = \begin{vmatrix} -26 & -7 & 12 \\ 11 & 3 & -5 \\ -5 & -1 & 2 \end{vmatrix}$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj } A$$

$$= \begin{vmatrix} -26 & -7 & 12 \\ 11 & 3 & -5 \\ -5 & -1 & 2 \end{vmatrix}$$

Now,

$$AX = B$$

$$\Rightarrow X = A^{-1}B = \begin{vmatrix} -26 & -7 & 12 \\ 11 & 3 & -5 \\ -5 & -1 & 2 \end{vmatrix} \begin{vmatrix} 0 \\ 7 \\ 4 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} x \\ y \\ z \end{vmatrix} = \begin{vmatrix} -1 \\ 1 \\ 1 \end{vmatrix}$$

$$\therefore x = -1, y = 1, z = 1$$

* Using matrix method solve the following equations:

$$x + y + z = 6$$

$$x - y + z = 2$$

$$2x + y - z = 1$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

$$|A| = 6 \neq 0$$

$$\text{adj } A = \begin{bmatrix} 0 & 3 & 3 \\ 2 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$$

$$X = A^{-1} B$$

$$= \frac{1}{|A|} \text{adj } A \cdot B$$

$$= \frac{1}{6} \begin{vmatrix} 0 & 4 & 2 \\ 18 & -6 & 0 \\ 18 & 2 & -2 \end{vmatrix} \Rightarrow \begin{vmatrix} 1 & 4 & 2 \\ 2 & -6 & 0 \\ 3 & 2 & -2 \end{vmatrix}$$

$$\therefore x = 1, y = 2, z = 3$$

• Inverse of a matrix :

Elementary row transformation method

find the inverse of $A = \begin{vmatrix} 3 & 0 & -3 & 4 \\ 2 & 0 & -3 & 4 \\ 0 & 1 & -1 & 1 \end{vmatrix}$

by elementary row transformation.

Q. Let $A = IA$

$$\Rightarrow \begin{vmatrix} 3 & 0 & -3 & 4 \\ 2 & 0 & -3 & 4 \\ 0 & 1 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{vmatrix} A$$

$$\Rightarrow \begin{vmatrix} 1 & 0 & 0 & 0 \\ 2 & 0 & -3 & 4 \\ 0 & 1 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{vmatrix} A, \quad R_1 \rightarrow R_1 - R_2$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & -3 & 4 & 0 \\ 0 & -1 & 1 & 2 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ -2 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] \quad A, R_2 = R_2 - 2R_1$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 8 \\ 0 & -3 & 4 & 0 \\ 0 & 0 & -1 & 1 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ -2 & 3 & 0 & 0 \\ -2 & -3 & 3 & 1 \end{array} \right] \quad A, R_3 = 3R_3 - R_2$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 6 & -9 & 12 & 0 \\ 2 & -3 & 3 & 0 \end{array} \right] \quad A, R_2 = R_2 + 4R_3$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ -2 & +3 & -4 & 0 \\ -2 & 3 & -3 & 0 \end{array} \right] \quad \begin{aligned} A, R_2 &= -\frac{1}{3}R_2, \\ R_3 &= -\frac{1}{3}R_3 \end{aligned}$$

$$\therefore A^{-1} = \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ -2 & +3 & -4 & 0 \\ -2 & 3 & -3 & 0 \end{array} \right]$$

Ans

Q * find the inverse of $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$
by elementary row transformation.

→ let,

$$A = IA$$

$$\Rightarrow \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} A, R_1 \leftrightarrow R_2$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -5 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -3 & 1 \end{bmatrix} A, R_3 \rightarrow R_3 - 3R_2$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ 5 & -3 & 1 \end{bmatrix} A, \begin{array}{l} R_1 \rightarrow R_1 - 2R_2 \\ R_3 \rightarrow R_3 + 5R_2 \end{array}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 5/2 \end{array} \right] = \left[\begin{array}{ccc|c} -2 & 1 & 0 & -2 \\ 1 & 0 & 0 & 1 \\ 5/2 & -3/2 & 1/2 & 1/2 \end{array} \right] \quad A, R_3 \rightarrow 1/2 R_3$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1/2 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 5/2 \end{array} \right] = \left[\begin{array}{ccc|c} 1/2 & -1/2 & 1/2 & 1/2 \\ -4 & 3 & -1 & -4 \\ 5/2 & 3/2 & -1/2 & 1/2 \end{array} \right] \quad \begin{array}{l} A, R_1 \rightarrow R_1 + R_3 \\ R_2 \Rightarrow R_2 - 2R_3 \end{array}$$

• System of Linear Equation:

* Homogeneous system of linear equation :-

A system of m linear equations in n variables $x_1, x_2, x_3, \dots, x_n$ of the following form is called non-homogeneous system of linear equations:

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

echelon form

Q * Test the consistency and solve if possible

$$x + 2y - z = 3$$

$$3x - y + 2z = 4$$

$$2x - 2y + 3z = 2$$

or,

$$x + 2y - z = 3$$

$$-7y + 5z = -5, L_2 \rightarrow L_2 - 3L_1$$

$$-6y + 5z = -4, L_3 \rightarrow L_3 - 2L_1$$

or,

$$x + 2y - z = 3$$

$$-7y + 5z = -5$$

$$L_2 : 3x - y + 2z = 4$$

$$3L_1 : 3x + 6y - 3z = 9$$

$$-7y + 5z = -5$$

$$L_3 : 2x - 2y + 3z = 2$$

$$2L_1 : 2x + 4y - 2z = 6$$

$$-6y + 5z = -4$$

$$L_3 \rightarrow 7L_3 - 6L_2$$

which is in echelon form,

[সমীকরণের সংখ্যা, চরকের সংখ্যা সমান হলে Unique solution পাওয়া যাবে]

So, there are 3 equations in 3 unknowns. So the given system has a unique solution.

$$x = 7/5, y = 1, z = 2/5$$

9 * Test the consistency and solve if possible

$$x + y + z = 6$$

$$x - y + z = 2$$

$$2x + y - z = 1$$

or,

$$x + y + z = 6$$

$$-2y = -4$$

$$\text{~~Equation 1~~}$$

$$-y - 3z = -11$$

$$L_2 \rightarrow L_2 - 4$$

$$L_3 \rightarrow L_3 - 2L_1$$

or,

$$x + y + z = 6$$

$$-2y = -4$$

$$-6z = -18$$

$$2L_3 : -2y - 6z = -22$$

$$L_2 : -2y = -4$$

$$-6z = -18$$

$$x=1, y=2, z=3$$

Q* Test the consistency and solve if possible

$$2x+3y-2z=5$$

$$x-2y+3z=2$$

$$4x-y+4z=1$$

or,

$$2x+3y-2z=5$$

$$-7y+8z=-1$$

$$L_2 \rightarrow 2L_2 - L_1$$

$$-7y+8z=-9$$

$$L_3 \rightarrow L_3 - 2L_1$$

or,

$$2x+3y-2z=5$$

$$-7y+8z=-1$$

$$0=-8, L_3 \rightarrow L_3 - L_2$$

which is in echelon form, and the last equation is of

the form $0=-8$, which is not true.

so, the given system is inconsistent and has no solution.

9 * Test the consistency and solve if possible

$$x + 5y + 4z - 13t = 3$$

$$3x - y + 2z + 5t = 2$$

$$2x + 2y + 3z - 4t = 1$$

or,

$$x + 5y + 4z - 13t = 3$$

$$-16y - 10z + 44t = -7, \quad L_2 \rightarrow L_2 - 3L_1$$

$$-8y - 5z + 22t = -5, \quad L_3 \rightarrow L_3 - 2L_1$$

or,

$$x + 5y + 4z - 13t = 3$$

$$-16y - 10z + 44t = -7$$

$$0 = -3$$

[inconsistent]

$$L_2 - 3L_1$$

$$3x - y + 2z + 5t = 2$$

$$3x + 15y + 12z - 39t = 9$$

$$-16y - 10z + 44t = -7$$

$$L_3 - 2L_1$$

$$2x + 2y + 3z - 4t = 1$$

$$2x + 10y + 8z - 26t = 6$$

$$-8y - 5z + 22t = -5$$

$$-16y - 10z + 44t = -10$$

$$-16y - 10z + 44t = -7$$

Q * Test the consistency and solve if possible

$$x + y + 2z = 0$$

$$y + z = 0$$

$$-2x + 3y + z = 0$$

or,

$$x + y + 2z = 0$$

$$y + z = 0$$

$$5y + 5z = 0 \quad L_3 \rightarrow L_3 + 2L_2$$

$$x + y + 2z = 0$$

$$y + z = 0$$

$$0 = 0 \quad L_3 \rightarrow L_3 - 5L_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

* সমীকরণের মা.খ্যা চলকের মা.খ্যা থেকে কম হলে একের
অধিক সমাধান পাওয়া যায়]

which is in echelon form and there are 2 equations in 3 unknowns. so the system has infinite number of solutions.

Here the free variable is z .

putting $z = -1$, we get $y = 1$ & $x = 1$

• Eigenvalue and Eigenvector of a matrix:

Let A be a $n \times n$ square matrix.

A column matrix $v \in \mathbb{R}^n$ is called an eigenvector of A if

$$Av = \lambda v$$

where λ is the eigenvalue of A .

characteristic matrix

of A is $\lambda I - A$

characteristic polynomial

of A is $|\lambda I - A|$

characteristic equation

of A is $|\lambda I - A| = 0$

Q * find all eigen value and eigen vector of

$$A = \begin{vmatrix} 1 & 4 \\ 2 & 3 \end{vmatrix}$$

, Then diagonalize it.

→ Characteristic matrix of A is, $\lambda I - A$

$$= \begin{vmatrix} \lambda - 1 & 0 \\ 0 & \lambda - 3 \end{vmatrix} = \begin{vmatrix} \lambda - 1 & -4 \\ -2 & \lambda - 3 \end{vmatrix}$$

$$= \begin{vmatrix} \lambda - 1 & 0 \\ 0 & \lambda - 3 \end{vmatrix} = \begin{vmatrix} \lambda - 1 & -4 \\ -2 & \lambda - 3 \end{vmatrix}$$

$$= \begin{vmatrix} \lambda - 1 & -4 \\ -2 & \lambda - 3 \end{vmatrix}$$

characteristic polynomial of A ,

$$|\lambda I - A|$$

$$= \begin{vmatrix} \lambda - 1 & -4 \\ -2 & \lambda - 3 \end{vmatrix}$$

$$= (\lambda - 1)(\lambda - 3) - 8 = \lambda^2 - 4\lambda - 5$$

$$= \lambda^2 - 4\lambda - 5$$

characteristic equation of A is,

$$|\lambda I - A| = 0$$

$$\Rightarrow \lambda^2 - 4\lambda - 5 = 0 \Rightarrow (\lambda + 1)(\lambda - 5) = 0$$

$$\therefore \lambda = -1, 5$$

$\therefore -1, 5$ are the equivalent of A .

Let, $v_1 = \begin{bmatrix} x \\ y \end{bmatrix}$ be the eigen vector corresponding to the eigenvalue $\lambda = -1$ such that,

$$(\lambda I - A) v_1 = 0$$

$$\Rightarrow \begin{vmatrix} -2 & -4 \\ -2 & -4 \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix} = 0$$

$$\Rightarrow -2x - 4y = 0$$

$$\Rightarrow -2x - 4y = 0$$

$$\Rightarrow -2x - 4y = 0$$

$$\Rightarrow x + 2y = 0$$

which is in echelon form. Here free variable

is, y putting $y = -1$, we get $x = 2$.

$\therefore \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ is eigenvector corresponding to

$$\lambda = -1.$$

Again,

Let $v_2 = \begin{bmatrix} x \\ y \end{bmatrix}$ be the eigen vector corresponding

to the eigenvalue $\lambda = 5$ such that,

$$(\lambda I - A) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{vmatrix} 4 & -4 \\ -2 & 2 \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix} = 0$$

$$\Rightarrow 4x - 4y = 0$$

$$\Rightarrow -2x + 2y = 0$$

$$\Rightarrow x - y = 0$$

$$\Rightarrow x - y = 0$$

$$\Rightarrow x - y = 0$$

which is in echelon form. Here free variable is y . Putting $y = 1$, we get $x = 1$.

$\therefore \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is the eigenvector corresponding to $\lambda = 5$.

Let,

$$P = \begin{pmatrix} -2 & 1 \\ 1 & 1 \end{pmatrix}$$

where P is a matrix whose column are the eigenvectors.

$$P^{-1} = ?$$

$$P = \begin{pmatrix} -2 & 1 \\ 1 & 1 \end{pmatrix}$$

$$V_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \quad V_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$|P| = \begin{vmatrix} -2 & 1 \\ 1 & 1 \end{vmatrix} = -2 - 1 = -3$$

$$0 = \begin{vmatrix} x & 1 \\ y & 1 \end{vmatrix} \Rightarrow \begin{vmatrix} x & 1 \\ y & 1 \end{vmatrix} = 0 \Rightarrow x - y = 0 \Rightarrow x = y$$

$$0 = P^T P - x P \Rightarrow$$

$$P_{11} = 1, P_{12} = -1, P_{21} = -1, P_{22} = -2$$

$$\text{adj. } P = \begin{pmatrix} 1 & -1 \\ -1 & -2 \end{pmatrix}^T = \begin{pmatrix} 1 & -1 \\ -1 & -2 \end{pmatrix}$$

$$P^{-1} = \frac{1}{-3} \begin{pmatrix} 1 & -1 \\ -1 & -2 \end{pmatrix}$$

Now,

$$B = P^{-1} A P$$

$$= \frac{1}{-3} \begin{pmatrix} 1 & -1 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ 1 & 1 \end{pmatrix}$$

$$= -\frac{1}{3} \begin{pmatrix} 1 & -1 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 2 & 5 \\ -1 & 5 \end{pmatrix}$$

$$= -\frac{1}{3} \begin{pmatrix} 3 & 0 \\ 0 & -15 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 5 \end{pmatrix}$$

which is a diagonal matrix where the diagonal elements are the eigenvalues of A.

Q * Diagonalize $A = \begin{vmatrix} 1 & 4 \\ 1 & 1 \end{vmatrix}$

characteristic matrix of A is, $\lambda I - A$

$$= \lambda \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - \begin{vmatrix} 1 & 4 \\ 1 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} - \begin{vmatrix} 1 & 4 \\ 1 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} \lambda - 1 & -4 \\ -1 & \lambda - 1 \end{vmatrix}$$

characteristic polynomial of A ,

$$|\lambda I - A|$$

$$= \begin{vmatrix} \lambda - 1 & -4 \\ -1 & \lambda - 1 \end{vmatrix}$$

$$= (\lambda - 1)(\lambda - 1) - 4$$

$$= \lambda^2 - \lambda - \lambda + 1 - 4$$

$$= \lambda^2 - 2\lambda - 3$$

characteristic equation of A is,

$$|\lambda I - A| = 0$$

$$\Rightarrow \lambda^2 - 2\lambda - 3 = 0$$

$$\Rightarrow \lambda^2 - 3\lambda + \lambda - 3 = 0$$

$$\Rightarrow \lambda(\lambda - 3) + 1(\lambda - 3) = 0$$

$$\Rightarrow (\lambda - 3)(\lambda + 1) = 0$$

$$\therefore \lambda = -1, 3.$$

$\therefore -1, 3$ are the equivalent of A .

Let,

$v_1 = \begin{bmatrix} x \\ y \end{bmatrix}$ be the eigen vector corresponding to the eigenvalue $\lambda = -1$ such that,

$$(\lambda I - A) v_1 = 0$$

$$\Rightarrow \begin{bmatrix} -2 & -4 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\Rightarrow -2x - 4y = 0$$

$$\Rightarrow -x - 2y = 0$$

$$\Rightarrow -2x - 4y = 0$$

$$\therefore x + 2y = 0$$

which is in echelon form the free variable is y putting $y = -1$, we get $x = 2$

$\therefore \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ is eigenvector corresponding to $\lambda = -1$.

Again,

let, $v_2 = \begin{pmatrix} x \\ y \end{pmatrix}$ be the eigen vector corresponding to the eigenvalue $\lambda = 3$ such that

$$(\lambda I - A)v_2 = 0$$

$$\Rightarrow \begin{pmatrix} 2 & -4 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$\text{or, } \Rightarrow 2x - 4y = 0$$

$$-x + 2y = 0$$

$$\text{or, } \Rightarrow 2x - 4y = 0$$

$$x - 2y = 0$$

which is in echelon form the free variable is y putting $y = 1$, we get $x = 2$

$\therefore \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ is the eigen vector corresponding to $\lambda = 3$.

Let,

$$P = \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix}$$

where P is a matrix

whose column are the eigenvectors.

$$|P| = \begin{vmatrix} 2 & 2 \\ -1 & 1 \end{vmatrix} = 2 + 2 = 4$$

$$P_{11} = 1, P_{12} = 1, P_{21} = -2, P_{22} = 2$$

$$\text{adj. } P = \begin{pmatrix} 1 & 1 \\ -2 & 2 \end{pmatrix}^T = \begin{pmatrix} 1 & -2 \\ 1 & 2 \end{pmatrix}$$

$$P^{-1} = \frac{1}{4} \begin{pmatrix} 1 & -2 \\ 1 & 2 \end{pmatrix}$$

Now,

$$B = P^{-1} A P$$

$$= \frac{1}{4} \begin{pmatrix} 1 & -2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix}$$

$$= \frac{1}{4} \begin{pmatrix} 1-2 & 4-2 \\ 1+2 & 4+2 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix}$$

$$= \frac{1}{4} \begin{pmatrix} -1 & 2 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix}$$

$$= \frac{1}{4} \begin{pmatrix} -4 & 0 \\ 0 & 12 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}$$

Q * Diagonalize :

$$i) A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$$

$$ii) A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

• Cayley Hamilton theorem :

Every square matrix satisfies its own characteristic equation.

Q * Verify the Cayley-Hamilton theorem and hence find A^{-1} for $A = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$

$\Rightarrow$ The characteristic matrix of A is, $\lambda I - A$

$$= \lambda \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} - \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} \lambda - 1 & -2 \\ -1 & \lambda - 1 \end{vmatrix}$$

The characteristic polynomial of A is, $\lambda^2 - 2\lambda - 1$

$$|\lambda I - A|$$

$$= \begin{vmatrix} \lambda - 1 & -2 \\ -1 & \lambda - 1 \end{vmatrix}$$

$$= (\lambda - 1)^2 - 2$$

$$= \lambda^2 - 2\lambda + 1 - 2$$

$$= \lambda^2 - 2\lambda - 1$$

The characteristic equation of A is $|\lambda I - A| = 0$

$$\Rightarrow \lambda^2 - 2\lambda - 1 = 0$$

$$\begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} - \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = \lambda I - A$$

In order to verify Cayley-Hamilton theorem, we have to show that,

$$A^2 - 2A - I = 0 \quad \text{--- (1)}$$

LHS :

$$A^2 - 2A - I$$

$$= \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} - 2 \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} - \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$$= \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} 2 & 4 \\ 2 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3-2-1 & 4-4-0 \\ 2-2-0 & 3-2-1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \text{R.H.S.}$$

$\therefore$ Hence Cayley-Hamilton theorem is verified.

Q * Verify the Cayley-Hamilton theorem and hence

find A^{-1} for $A = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$

Multiplying ① by A^{-1} on both sides we get,

$$A^2 A^{-1} - 2AA^{-1} - I \cdot A^{-1} = 0$$

$$\Rightarrow A - I - A^{-1} = 0$$

$$\Rightarrow A^{-1} = A - I = \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} - \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 2 \\ 1 & 0 \end{vmatrix}$$

Q * Verify the Cayley-Hamilton theorem and hence

find A^{-1} for

i) $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$

ii) $A = \begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix}$

iii) $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 4 \end{pmatrix}$

Q * Determine which of the following are the eigenvectors

for $A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$

showing your analysis graphically;

a) $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

c) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

e) $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$

b) $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

d) $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$

f) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

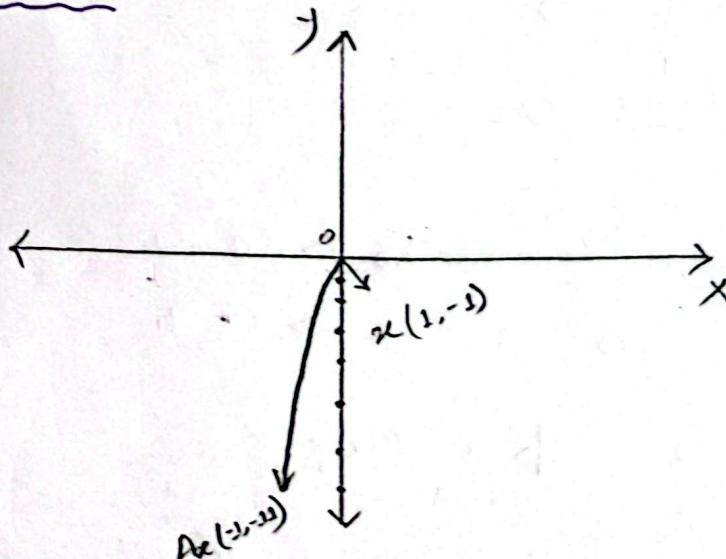
• a) $A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$, $x = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$Ax = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 \\ -1 \end{pmatrix} = (-1) \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Since $Ax = \lambda x$ is not satisfied, so $x = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is not an eigenvector for A .

Graphical Test:



the direction of Ax
and x are not same.
So, x is not an
eigenvector for A .

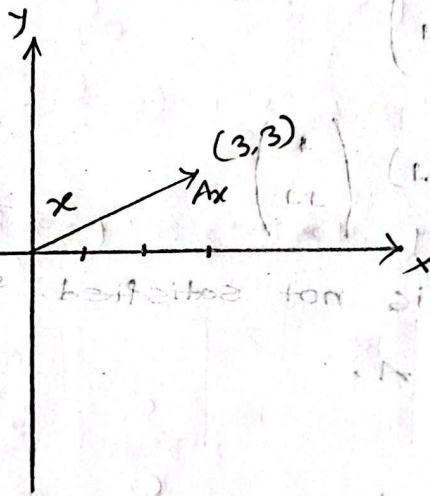
• f) $A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$, $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$Ax = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Since $Ax = \lambda x$ is satisfied. So $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigen-vector for A .

Graphical Test :



Since the direction of Ax and x are same, so x is eigen-vector for A .

• Linear Dependence :

The set of vectors $x_1, x_2, \dots, x_n$ is said to be linearly dependent if there exists scalars $a_1, a_2, \dots, a_n$ not all zero such that,

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = 0$$

Q * Verify the set of vectors $(2, -1, 4)$, $(3, 6, 2)$ and $(2, 10, -4)$ are linearly dependent or not.

→ Set the linear combination of the given vectors to zero vector using scalars x, y, z . we get,

$$x(2, -1, 4) + y(3, 6, 2) + z(2, 10, -4) = 0$$

$$\Rightarrow (2x + 3y + 2z, -x + 6y + 10z, 4x + 6y - 4z) = (0, 0, 0)$$

then we get,

$$2x + 3y + 2z = 0$$

$$-x + 6y + 10z = 0$$

$$4x + 6y - 4z = 0$$

$$\Rightarrow 2x + 3y + 2z = 0$$

$$15y + 22z = 0$$

$$-8z = 0$$

$$L_2 \rightarrow 2L_2 + L_1$$

$$L_3 \rightarrow L_3 - 2L_1$$

which is in echelon form there are three equations in three unknowns. So the system has zero solution that is $x=0, y=0, z=0$. the given vectors are linearly

independent.

$\{0, 2, 0\}$ are independent
non-0 are dependent

Q * Verify the set of vectors.

$(1, 1, 0)$, $(0, -1, 1)$ and $(2, 0, -1)$ are linearly dependent or not.

Q * Show that the vectors $(1, 1, 1)$, $(1, 2, 3)$, $(2, 3, 5)$ are linearly independent.

→ Set the linear combination of the given vectors to zero vector using scalars x, y, z . we get,

$$x(1, 1, 1) + y(1, 2, 3) + z(2, 3, 5) = 0$$

$$\Rightarrow (x+y+2z, x+2y+3z, x+3y+5z) = (0, 0, 0)$$

then we get,

$$x + y + 2z = 0$$

$$x + 2y + 3z = 0$$

$$x + 3y + 5z = 0$$

$$\Rightarrow x + y + 2z = 0$$

$$y + 2z = 0$$

$$z = 0 \quad L_3 \leftarrow L_3 - 2L_2$$

which is the echelon form & there are three

equations is three unknown, so the system has zero solution that is $x=0, y=0, z=0$, so the given vectors are linearly independent.

Q * Show that the vectors $(1, -2, 3), (5, 6, -1), (3, 2, 1)$ are linearly dependent.

→ Set the linear combination of the given vectors to zero vector using scalars x, y, z , we get,

$$x(1, -2, 3) + y(5, 6, -1) + z(3, 2, 1) = 0$$

$$\Rightarrow (x+5y+3z, -2x+6y+2z, 3x-y+z) = 0$$

then we get,

$$x + 5y + 3z = 0$$

$$-2x + 6y + 2z = 0$$

$$3x - y + z = 0$$

$$\Rightarrow x + 5y + 3z = 0$$

$$16y + 8z = 0$$

$$-16y - 8z = 0$$

$$\Rightarrow x + 5y + 3z = 0$$

$$16y + 8z = 0$$

$$0 = 0, L_3 \rightarrow L_3 + L_2$$

which is in echelon form and there are 2 equations in unknowns. So the system has non-zero solution. So the given vectors are linearly dependent.

Q * Show that the vectors $(1, 1, 2)$, $(1, 0, 1)$, $(2, 1, 3)$ are linearly dependent.

• Rank of a matrix:

A positive number r is said to be the rank of matrix

A if the matrix A satisfies the following condition:-

① there exists at least one non-zero minor of order r

② Every minor of order $(r+1)$ & higher if any vanishes.

Q * find the rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$

let,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$$

$$|A|_2 = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{vmatrix} = 0$$

So we have to take 2 rowed minor of A , say,

$$\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} \rightarrow (2 \times 2)$$

$$\Rightarrow \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} = 3 - 4 = -1 \neq 0$$

$$\therefore \rho(A) = 2$$

Q * find the rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$

• Normal form of a matrix:

The normal form of a matrix of rank r is one of the following form,

$$I_r, \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} = [I_r, 0] \cdot \begin{bmatrix} I_r \\ 0 \end{bmatrix}$$

where I_r is an identity matrix of order r .

$$\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix}$$

Q * Reduce the matrix A to its normal (canonical) form and hence find the rank of A, where

$$A = \begin{vmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{vmatrix}$$

$$\Rightarrow A = \begin{vmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{vmatrix}$$

$$\sim \begin{vmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{vmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\sim \begin{vmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{vmatrix}$$

$$R_3 \rightarrow R_3 - R_1$$

$$R_4 \rightarrow R_4 - R_1$$

$$\sim \left| \begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & & \\ 0 & 1 & -3 & -1 & & \\ 0 & 0 & 1 & 0 & 0 & \\ 0 & 0 & 0 & 0 & 0 & \end{array} \right| \begin{array}{l} R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2 \end{array}$$

$$\sim \left| \begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & \\ 0 & 1 & -3 & -1 & & \\ 0 & 0 & 1 & 0 & 0 & \\ 0 & 0 & 0 & 0 & 0 & \end{array} \right| \begin{array}{l} C_3 \rightarrow C_3 - C_1 \\ C_4 \rightarrow C_4 - C_1 \end{array}$$

$$\sim \left| \begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & \\ 0 & 1 & 0 & 0 & 0 & \\ 0 & 0 & 1 & 0 & 0 & \\ 0 & 0 & 0 & 0 & 0 & \end{array} \right| \begin{array}{l} C_3 \rightarrow C_3 + 3C_2 \\ C_4 \rightarrow C_4 + C_2 \end{array}$$

$$\sim \left| \begin{array}{ccc|ccc} I_2 & 0 & & & & \\ 0 & 0 & & & & \end{array} \right|$$

$$\rho(P(A)) = 2$$

* Reduce the matrix A to its normal (canonical) form and hence find the rank of A , where,

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$$

বি:দ্র : ভুল এটি থাকলে ক্ষমাসুলভ দৃষ্টিতে দেখবেন ধন্যবাদ 😊